

Course Project

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Project Ideas

Choose any one of the following six projects. All projects are designed to cover the same core syllabus topics so grading remains consistent regardless of which project a team selects. (Team members **max 5 students**)

1. Medical Image Diagnosis System

Domain	Healthcare / Medical Imaging
Dataset	Chest X-Ray Images (NIH or Kaggle), Skin Lesion ISIC dataset
Task	Multi-class disease classification from medical images with explainability
Phase 1 — Data & Classical ML	EDA on image metadata & labels. Apply KNN and Naïve Bayes on extracted hand-crafted features (HOG, LBP). Use PCA/LDA for dimensionality reduction. Apply Ensemble methods (Random Forest, Gradient Boosting). Evaluate with metrics, visualize confusion matrices.
Phase 2 — Deep Learning (CNN/AE/Transfer)	Build CNN from scratch for image classification. Compare optimizers (SGD, Adam, RMSprop) and tune hyperparameters. Add Autoencoder for anomaly/outlier detection. Apply Transfer Learning (VGG, ResNet, EfficientNet) and compare results.
Phase 3 — Advanced (Transformers/GANs/XAI)	Integrate an LSTM/GRU over temporal patient data (if multi-visit). Use Vision Transformer (ViT) for classification. Generate synthetic medical images using GAN/VAE to handle class imbalance. Apply XAI techniques (Grad-CAM, LIME, SHAP) to explain model decisions.

2. Sentiment & Fake News Detection Pipeline

Domain	NLP / Social Media Analytics
Dataset	Twitter Sentiment140, LIAR dataset for fake news, Amazon Reviews
Task	Detect sentiment polarity and classify news authenticity
Phase 1 — Data & Classical ML	Text EDA: word clouds, frequency plots. TF-IDF + KNN for sentiment classification. Naïve Bayes classifier (natural fit for text). Dimensionality reduction (TruncatedSVD/LSA). Ensemble: Random Forest, XGBoost on TF-IDF features. Tune hyperparameters (n-estimators, max depth).
Phase 2 — Deep Learning	Train a 1D-CNN text classifier on word embeddings. Build an Autoencoder

(CNN/AE/Transfer)	for anomalous text detection. Use Transfer Learning with pretrained word embeddings (GloVe, FastText). Experiment with different optimizers and dropout for regularization.
Phase 3 — Advanced (Transformers/GANs/XAI)	Build LSTM/Bi-LSTM for sequence classification. Implement an Attention mechanism to highlight key words. Fine-tune BERT/DistilBERT (Transformer). Apply SHAP and LIME for interpretability. Conditional GAN to generate adversarial fake news samples.

3. Autonomous Vehicle Perception Module

Domain	Computer Vision / Autonomous Systems
Dataset	CIFAR-10 (traffic signs), German Traffic Sign Dataset, BDD100K subset
Task	Object detection and road scene classification for self-driving cars
Phase 1 — Data & Classical ML	Analyze traffic sign dataset: class balance, pixel statistics. Feature extraction + KNN & Naïve Bayes baseline. PCA for compression, measure accuracy trade-offs. Ensemble classifiers (Bagging, Boosting). Grid search for hyperparameter optimization.
Phase 2 — Deep Learning (CNN/AE/Transfer)	Deep CNN for traffic sign recognition. AutoEncoder for noisy image reconstruction. Transfer Learning: fine-tune MobileNet/VGG for in-car camera data. Compare optimizers and learning rate schedulers. Visualize CNN feature maps.
Phase 3 — Advanced (Transformers/GANs/XAI)	Add LSTM for processing video frame sequences. Implement Attention to focus on relevant scene regions. Generate synthetic traffic scenarios using DCGAN. Apply GradCAM and SHAP to explain predictions. Document model fairness across weather/lighting conditions.

4. Financial Market Forecasting & Anomaly Detection

Domain	Finance / Time-Series
Dataset	Yahoo Finance stock data, Credit Card Fraud (Kaggle), Crypto market data
Task	Predict stock trends and detect fraudulent transactions
Phase 1 — Data & Classical ML	EDA: correlation matrices, rolling statistics. Feature engineering (RSI, MACD). KNN regressor for price prediction. Naïve Bayes for buy/sell signal classification. Dimensionality reduction with PCA. Ensemble regressors (XGBoost, Random Forest) with hyperparameter tuning.
Phase 2 — Deep Learning (CNN/AE/Transfer)	1D-CNN for pattern recognition in time series. AutoEncoder for fraud/anomaly detection (unsupervised). Transfer Learning: adapt model trained on one stock to others. Tune learning rate, batch size, regularization. Evaluate detection thresholds with ROC-AUC.
Phase 3 — Advanced (Transformers/GANs/XAI)	LSTM / GRU for multi-step financial forecasting. Attention-based sequence model to identify key time steps. GAN for synthetic financial data augmentation. Transformer for long-range dependency modeling. SHAP to

explain which features drive predictions.

5. Smart Agriculture Crop Disease Detection

Domain	Agriculture / Remote Sensing
Dataset	PlantVillage dataset (Kaggle), Crop type satellite imagery
Task	Classify plant diseases from leaf images and recommend treatments
Phase 1 — Data & Classical ML	Exploratory analysis of leaf images and disease labels. Color histogram + texture features + KNN/Naïve Bayes. PCA on image features for visualization. Ensemble methods (Random Forest, AdaBoost). Cross-validation and hyperparameter grid search.
Phase 2 — Deep Learning (CNN/AE/Transfer)	Train CNN on PlantVillage. Autoencoder for healthy vs. diseased leaf anomaly detection. Fine-tune EfficientNet / MobileNetV2 via Transfer Learning. Experiment with data augmentation and optimizers. Build a confusion matrix dashboard.
Phase 3 — Advanced (Transformers/GANs/XAI)	LSTM over time-series crop health indicators. Attention for multi-scale leaf feature focus. DCGAN to synthesize rare disease samples for data augmentation. Vision Transformer (ViT) for classification. LIME/Grad-CAM for explainable disease localization.

6. Human Activity Recognition & Health Monitoring

Domain	Healthcare / IoT / Wearables
Dataset	UCI HAR Dataset (smartphone accelerometer/gyroscope), WISDM dataset
Task	Classify human activities (walking, sitting, running) from sensor signals
Phase 1 — Data & Classical ML	EDA on sensor signals: time-domain features (mean, std, FFT). KNN and Naïve Bayes on handcrafted features. PCA for sensor dimensionality reduction. Ensemble: Random Forest vs. Gradient Boosting. Hyperparameter tuning and cross-subject evaluation.
Phase 2 — Deep Learning (CNN/AE/Transfer)	1D-CNN for raw sensor classification. AutoEncoder for detecting irregular/unhealthy activity patterns. Transfer Learning from one subject's model to another. Optimizer comparison (Adam vs. SGD with momentum). Evaluate with per-class F1 score.
Phase 3 — Advanced (Transformers/GANs/XAI)	LSTM / Bi-LSTM for temporal activity sequence modeling. Attention mechanism to weight informative time steps. Conditional GAN to synthesize minority class activity data. Transformer encoder for activity recognition. SHAP for feature importance across sensor channels.

Deliverables per Phase

Phase 1 Deliverables

- Jupyter Notebook with
 1. Full EDA (plots, statistics, insights)
 2. Implementation of KNN and Naïve Bayes with performance metrics (accuracy, F1, ROC)
 3. Dimensionality reduction visualization (PCA / LDA explained variance plots)
- Brief written report (PDF) — max 5 pages

Phase 2 Deliverables

- CNN architecture diagram and training curves (loss, accuracy)
- Autoencoder reconstruction visualizations and anomaly detection results
- Transfer Learning: comparison table of pretrained models and fine-tuned results
- Optimizer comparison experiment (at least 2 optimizers, tracked metrics)
- Updated report with Phase 2 results appended (cumulative, max 10 pages)

Phase 3 Deliverables

- LSTM/Transformer model with performance vs. Phase 2 CNN comparison
- Attention visualization (heatmaps or weight plots)
- GAN/VAE: generated sample gallery and FID or qualitative evaluation
- XAI section: SHAP summary plot, LIME explanations, or Grad-CAM overlays
- Final comprehensive report (max 20 pages).

Evaluation Criteria

Criterion	Description	Weight
Coverage of Syllabus Topics	All required methods for the phase must be implemented and compared	35%
Code Quality & Reproducibility	Clean, documented, runnable code in a structured notebook/repo	20%
Analytical Depth	Meaningful experiments, ablations, and insight-driven discussion	25%
Report Quality	Clear writing, proper figures, tables, and citations	10%
Presentation / Demo	Phase 3 only — clarity of demo, ability to answer questions	10%